Satyam Awasthi

Researcher & Software Engineer

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Professional Summary

Computer Science researcher specializing in the intersection of **Autonomous Decision-Making**, **Spatial Perception**, and **Robot Learning** within Embodied Systems, with a core focus on engineering verifiably safe, constraint-bound execution loops.

Education

University of California, Santa Barbara (UCSB) — Santa Barbara, CA
M.S. in Computer Science (GPA: 3.91/4.00) — Sep 2021 – Jun 2023
Graduate Research at the Four Eyes Lab focusing on calibrating eye-tracking frameworks for mixed-reality locomotion and multi-modal spatial perception.
Advisor: Prof. Tobias Höllerer · Co-Advisor: Prof. Michael Beyeler

Indian Institute of Technology, Kharagpur (IIT Kharagpur) — Kharagpur, India
B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0) — Jul 2016 – Jul 2020
Research focused on 2D kinematic profiles, decentralized multi-agent motion planning, and hardware-agnostic stacks for Sim2Real transfer.

Skills

Programming & Simulation: Python, C++, MATLAB, Kotlin, Java, Gazebo, Rviz, Unity, Git
Robotics, Vision & AI Frameworks: ROS/ROS2, OpenCV, PyTorch, LangChain, Pydantic, ARCore/iOS ARKit
Core Methodologies: State Estimation (EKF), Control Theory (MPC), Kinematic Modeling, Spatial Perception
Fields of Interest: Autonomous Decision-Making, Spatial Perception, Robot Learning

Publications

Refereed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing
Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.
Proceedings of the 14th International Conference on COMmunication Systems & NETworkS (COMSNETS), 2022.
DOI: 10.1109/COMSNETS53615.2022.9668531
Engineered an adaptive coordinate transformation pipeline using gravitational alignment and PUPD micro-gesture filtering to resolve drift-free, 2D kinematic profiles of wall-writing from wearable inertial sensors.

Refereed Poster Papers

Eye Tracking Performance in Mobile Mixed Reality
Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias Höllerer.
IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), 2024.
DOI: 10.1109/VRW62533.2024.00321
Expanded the tracking framework to benchmark low-latency performance profiles and gaze errors via a robust 3-device user study across diverse mixed-reality locomotion layouts. Supported by NSF Grant #2211784.

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion
Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias Höllerer.
IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), 2023.
DOI: 10.1109/ISMAR-Adjunct60411.2023.00104
Developed a novel eye-tracking test suite (EyeTTS) and engineered unique software calibration pipelines optimized to correct tracking drift during active human locomotion; validated via a 2-device user study. Supported by NSF Grant #2211784.

Experience

Intuit Inc Mountain View, CA · Mar 2024 – Jan 2026
Software Engineer 2

- Built dual-stage **agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc Mountain View, CA · Apr 2023 – Nov 2023
Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.
- Designed a dynamic, state-driven carousel UI framework optimized for client-side rendering efficiency.

Coupang Global LLC Mountain View, CA · Jun 2022 – Sep 2022
Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute Noida, India · Sep 2020 – Sep 2021
Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Projects

JitterNot Jan 2022 – Mar 2022
LSTM-Based Model Predictive Controller for Adaptive Video Streaming UCSB

- Developed an LSTM network for real-time throughput prediction, transforming the closed-loop control system from MISO to SISO.
- Implemented an MPC controller leveraging Receding Horizon Control (RHC) principles to dynamically calculate optimal video resolution.

Autonomous Ground Vehicle & Swarm Robotics Jan 2016 – Dec 2018
End-to-End Perception, Planning, and Multi-Agent Navigation Stack IIT Kharagpur

- Developed a hardware-agnostic navigation stack in ROS, executing an Extended Kalman Filter (EKF) to fuse multi-modal sensor streams (LiDAR, IMU, GPS) for zero-shot **Sim2Real transfer**.
- Implemented real-time visual obstacle segmentation using a custom U-Net architecture alongside a hybrid motion planner combining global (A^*/Dijkstra) paths with a Dynamic Window Approach (DWA) for local collision avoidance.
- Formulated a decentralized swarm coordination network utilizing peer-to-peer relative pose estimation via AprilTags over an ad-hoc XBee mesh topology.

In Motion — HAR using DeepConvLSTM Nov 2019 – Nov 2019
Distributed Mobile Computing System for Human Activity Recognition IIT Kharagpur

- Designed a client-server telemetry pipeline streaming high-frequency IMU sensor arrays for rolling window inference.
- Implemented a DeepConvLSTM core architecture using TimeDistributed CNN layers to model spatial-temporal features efficiently.

CohesiveAR Apr 2022 – Jun 2022
Augmented Reality Pipeline for Texture Extraction and Homography Mapping UCSB

- Developed a spatial perception pipeline using Google ARCore and OpenCV to map and project dynamic 3D scene geometry into unified 2D reference frames.
- Implemented matrix homography and perspective rectification algorithms to correct for camera tilt skew, extracting invariant surface textures from variable viewpoints.

Tweeter: Distributed Infrastructure Testbed Sep 2021 – Dec 2021
High-Concurrency Microblogging Testbed on AWS EC2 Clusters UCSB

- Conducted high-throughput load testing up to 512 users/sec, mapping user trajectories as state machines to latency scaling curves.

- Optimized read throughput by deploying nested fragment caches and database query preloading to eliminate bottleneck points.

Teaching

University of California, Santa Barbara — Santa Barbara, CA
Graduate Teaching Assistant — Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Academic Service & Peer Review

Technical Program Committee & Peer Reviewer
ACM Symposium on User Interface Software and Technology (UIST) — May 2024
Served as an external peer reviewer evaluating full paper submissions for novel interface hardware, interactive system architectures, and spatial tracking methodologies.